Developing a Surrogate Model for Underground Hydrogen Storage using Deep Learning.

Report

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MSc Geo-Energy with Machine Learning and Data Science

https://github.com/ese-ada-lovelace-2025/irp-tlp25

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1 AI Acknowledgements

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2 Abstract

In this study, we address the problem of constant injection, withdrawal and idle cycle lengths, and operational parameters currently used in hydrogen storage surrogate models. Deep learning surrogate models have been a growing area of subsurface modelling research in recent years. However, surrogate models require huge amounts of compute capacity to cover a large range of uncertainty which may not be needed. Underground hydrogen storage, in depleted gas fields, is well-placed to use compute efficiently. Due to its cyclical operating model, similar scenarios occur on a yearly basis. Current surrogate models for underground hydrogen storage have been developed using constant cycle lengths and two dimensional data. For future operational use of surrogate models, they must assess a varying range of cycle lengths and operational parameters. This is due to it being thought that hydrogen storage will be operated in the same way as natural gas storage. We show that natural gas storage cycle lengths are variable. We present a highly accurate three dimensional underground hydrogen storage surrogate model, using a U-Net architecture. Hydrogen saturation plume and pressure have maximum errors of 1.6% and 0.26% respectively. Predictions of injection rate and hydrogen stored have errors of 5.3% and 0.649%. The findings in this study create a framework for future underground hydrogen storage sites to develop highly accurate surrogate models which will enable operational optimisation and integration into global energy systems.

3 Introduction

The application of deep learning to subsurface modelling and reservoir simulation is a growing area of research with a high potential impact. By developing surrogate models using deep learning, simulations can be completed in considerably less time than conventional simulators that use numerical methods, once they are trained. In carbon capture and storage, surrogate modelling resulted in flow predictions being nearly 700,000 times quicker than existing methods (Wen et al., 2023). Research in geothermal and oil and gas simulations have shown similar results, reducing computational time by a factor of 100 and 1000 respectively (Deng et al., 2025; Huang et al., 2023). These examples show the huge potential of using surrogate models, developed using deep learning, to reduce the computational time of reservoir simulations which enables rapid screening of potential sites and optimise operations.

The storage of hydrogen is thought to play an important role in the transition to net zero, allowing seasonal energy storage at a grid scale (Krevor et al., 2023). Excess energy produced by renewable sources, such as solar and wind, can be used to produce hydrogen through the electrolysis of water (Kumar and Lim, 2022). This reduces the need for renewable energy curtailment, which led to the total curtailment of 10TWh of electricity in Great Britain alone during 2025 (Montel,

2026). Not only is energy lost, but developers lose revenue, leading to extra costs for governments through subsidies. These secondary impacts can make investing in renewables, which is crucial for reaching net zero, unattractive for both private investment and government support (IEA, 2025). Underground hydrogen storage in depleted gas fields is an area of growing interest. This is due to the higher storage capacity compared to salt caverns, strong-understanding of the reservoir geology and low initial capital expenditure (Fu et al., 2025).

In studies focused on developing hydrogen storage surrogate models, synthetic datasets are generated due to the lack of active sites in the world. These datasets are then used to develop a surrogate model. These methods have proven to be highly successful at predicting hydrogen migration and reservoir plume evolution. Fu et al. (2025) achieved high accuracy, achieving R^2 values of 0.996 and 0.991 for hydrogen saturation and pressure respectively. Mao et al. (2024) investigated the effects of cushion gas and developed a surrogate model. Hydrogen withdrawal efficiency, hydrogen purity, gas-water ratio and well injectivity were predicted with high accuracy. The results highlighted that cushion gas has negligible impacts on the performance metrics and storage performance improved with the number of cycles. In addition, the surrogate model allowed for the key performance metrics to be predicted five million times quicker than traditional physics simulators.

Other approaches have focussed specifically on operational parameters. Du et al. (2025) focussed on predicting hydrogen mass stored, gaseous hydrogen mass, hydrogen production rate and hydrogen dissolution rate. The surrogate model developed, only used the input rate to predict the focussed metrics. The study showed that long short-term memory models are the most effective for handling non-linear dynamic changes, which is most relevant for predicting production rate.

Although surrogate models allow for rapid prediction of saturation, pressure buildup and operational parameters, in areas such as carbon capture and storage and oil and gas production this is largely unnecessary. Both these fields are non-cyclical and therefore the models are often single-use. They are not designed to be continuously used or scaled. Unless the models developed can generalise with high accuracy to multiple reservoir sizes, heterogeneities and operational parameters they have limited use (Motaei et al., 2026).

In systems which are cyclical like geothermal energy and gas storage, the use of surrogate models presents an area of research with a high potential impact. Due to their cyclical nature, similar scenarios can repeat multiple times through a storage sites life cycle. The primary issue is that hydrogen storage surrogate models currently use constant cycle lengths. In the studies by Fu et al. (2025) and Mao et al. (2024), one cycle consisted of six months of injection and six months of withdrawal. Du et al. (2025) define a cycle as a five-month phase of hydrogen injection, a one-month shut-in phase and six-month production phase. Constant cycle structure is a limitation of the current studies (Mao et al., 2025). Underground hydrogen storage is thought to operate in the same way as underground natural gas storage (Tarkowski, 2019). The need for it derives from renewable energy curtailment. Existing natural gas storage sites do not operate with a constant cycle length and renewable energy curtailment varies seasonally (Pellow et al., 2015). Surrogate models for hydrogen storage must assess varying cycle lengths and rates of withdrawal and injection. Current hydrogen storage surrogate models also focus on aiming to generalise across multiple reservoir sizes and heterogeneities. For hydrogen storage in depleted gas fields this is largely unnecessary. Many of the sites that have been investigated for underground hydrogen storage (e.g. Roden, Viking A and Rough gas fields) have well-documented subsurface and formation understanding (Saadat et al., 2024). As a consequence, surrogate models do not need to assess a large range of uncertainty in the heterogeneity of the fields; instead they should focus on operational parameters.

In this study we demonstrate the use of surrogate modelling for hydrogen storage. We focus on assessing varying cycle lengths and operational rates which have been noted as an area for future work (Mao et al., 2025). The cycle lengths and rates are generated using European natural gas storage sites data as an analogue. We predict injection and withdrawal rates, hydrogen stored, hydrogen saturation and pressure. We show our three dimensional surrogate model, developed using a U-Net architecture, to be highly accurate for predicting some of the key variables in hydrogen storage.

4 Methodology

4.1 Baseline Model Description

The reservoir model used to generate the hydrogen storage dataset is adapted from the research by Hogeweg et al. (2022). The reservoir model is a grid with a semi-artificial geological structure. The reservoir's porosity and permeability are distributed heterogenously and designed to represent sandstone formations found in Northern Germany. It is discretised into a 61 x 61 x 12 grid. Each individual grid block is 50m in the x and y directions with varying height. Hogeweg et al. (2022) use two schedules in their study; a conversion from a natural gas storage site to underground hydrogen storage and regular operations. In this study, the conversion schedule is kept initially and the operational schedule is adapted, which is described later.

4.2 Natural Gas Data

The operational schedules in this study use data from existing natural gas storage sites as an analogue for the hydrogen storage schedules. The data was collected from the Aggregated Gas Storage Inventory, using historical data until 08/06/2026. The data was collected from 29 storage sites from across Europe (GIE, 2026). The monthly mean injection and withdrawal was calculated for each site. This was the most robust method; averaging over smaller periods led to longer-term trends being broken up by outlier days. Any cycle lengths with less than two occurrences were dropped. The different cycle types, injection, withdrawal and idle, were then normalised between zero and one using min-max normalisation. The injection and withdrawal rates were normalised together using min-max normalisation. This was done to keep the relative relationship between them.

Tests were done to assess any correlations which would determine the schedule generation method. There was no correlation between cycle length and subsequent cycle length and no correlation between the injection or withdrawal rate and the stage in the relevant cycle. The rates were split into 3 groups for both injection and withdrawal cycle types, giving a total of 6. The 3 groups were based on the normalised cycle position with the splits representing the start, middle and end of an injection or withdrawal cycle. Although no correlation was found, the distribution of the split groups was statistically different (Table 1).

4.3 Dataset Generation

To generate the hydrogen storage cycle lengths, latin hypercube sampling was used to generate samples across a uniform distribution. This was then transformed to match the distribution of the normalised injection or withdrawal cycle length distribution. The same method was applied to

Cycle Position	Mean	Median	STD	$0.33 < x \leq 0.66$		$0.66 < x \leq 1.0$	
				KS Stat	P-Value	KS Stat	P-Value
Injection							
$0 \leq x \leq 0.33$	0.696	0.680	0.129	X	X	X	X
$0.33 < x \leq 0.66$	0.766	0.770	0.138	0.234	0.00	X	X
$0.66 < x \leq 1.0$	0.709	0.697	0.138	0.0699	0.0869	0.205	0.00
Withdrawal							
$0 \leq x \leq 0.33$	0.425	0.434	0.157	X	X	X	X
$0.33 < x \leq 0.66$	0.308	0.297	0.183	0.308	0.00	X	X
$0.66 < x \leq 1.0$	0.382	0.389	0.176	0.14	0.00	0.184	0.00

Table 1: Summary statistics and Kolmogorov-Smirnov (K-S) test results indicating statistically different distribution. Note the start and end injection distributions were not statistically different. However, for the purpose of generating a hydrogen storage dataset that emulated the natural gas cycles, this would not have a significant impact.

generate the idle cycle lengths. The number of idle samples to generate was calculated by finding the probability of an idle cycle after an injection or withdrawal cycle.

The same method of generating samples was used to get the injection and withdrawal rate values. This method was used for the 6 rate distributions defined from the natural gas data. The number of samples generated for each distribution matched the number of corresponding points in the synthetic cycle lengths dataset. The generated rates were added to the synthetic injection and withdrawal cycles.

The hydrogen storage dataset had 1043 individual sites. Each site had a minimum 120 months. For this study one individual point represents one month, defined as 30 days. A maximum rate of injection and withdrawal was defined as the rate used in the baseline model by Hogeweg et al. (2022). This maximum rate was multiplied by the synthetic rates. Wellbore hole pressure limits were added to avoid overpressure and leakage of injected hydrogen. The maximum wellbore hole pressure was set at 102 bar and the minimum was set at 96.5 bar. These values are the limits seen during the regular operating schedule in the baseline model. For each month an Eclipse script was generated incorporating these values and limits, the individual month scripts were then concatenated to create one script per site. The 1043 site scripts were then ran on Eclipse E300 and the fluid injection composition was 100% hydrogen.

4.4 Preprocessing of Surrogate Model Inputs

4.4.1 Processing of Eclipse Simulation Results

For each site the dataset was cut to 120 timesteps from after the baseline model transition from natural gas storage to underground hydrogen storage. The first output of each month was selected. This is necessary due to Eclipse outputting a varying amount of smaller timesteps for each inputted month. For the individual grid cell data, the hydrogen saturation and pressure were selected.

Hydrogen saturation was calculated as the gas saturation multiplied by the vapour component mole fraction of hydrogen. The corresponding summary file of scalar outputs was then selected.

4.4.2 Model Inputs and Outputs

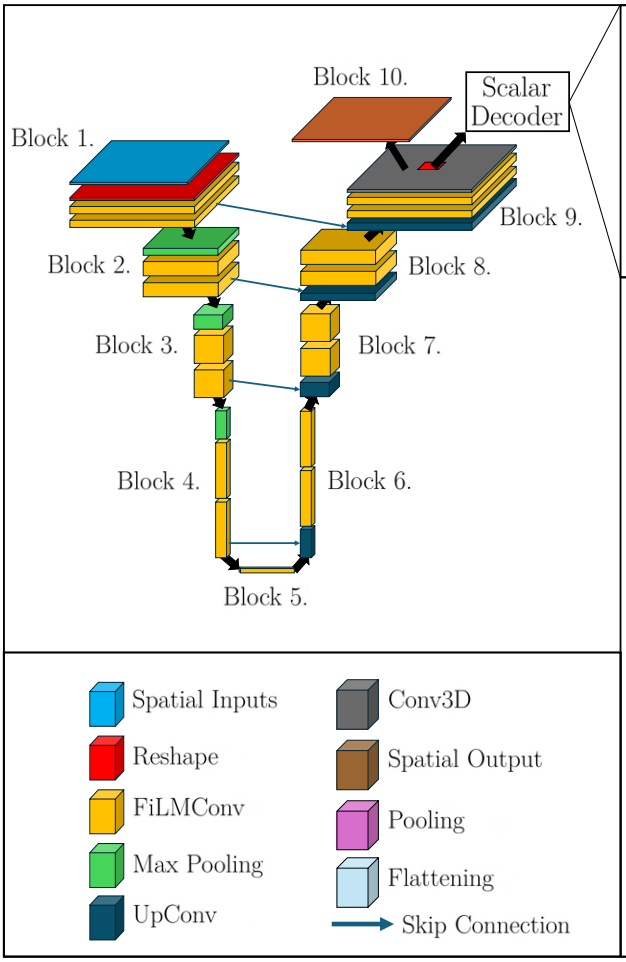
Two spatial components and seven scalar components were inputted to the model. The model then outputted two spatial components and 3 scalar components (Table 2). This resulted in 124,117 input and target pairs. These were split into train, validation and test groups. A random split of 0.7, 0.15 and 0.15 was used for the groups respectively. The spatial inputs were inputted once at the top of the model. The scalar inputs were inputted to the model within each FiLMConv Layer (fig. 1).

Variable	Time	Type
Input		
H_2 Saturation	t	Spatial
Pressure	t	Spatial
Eclipse Input Injection Rate	t	Scalar
Eclipse Input Withdrawal Rate	t	Scalar
Injection Wellbore Hole Pressure	t	Scalar
Withdrawal Wellbore Hole Pressure	t	Scalar
Idle Wellbore Hole Pressure	t	Scalar
Hydrogen Stored	t	Scalar
Month	t	Scalar
Output		
H_2 Saturation	t+1	Spatial
Pressure	t+1	Spatial
Eclipse Output Injection Rate	t+1	Scalar
Eclipse Output Withdrawal Rate	t+1	Scalar
Hydrogen Stored	t+1	Scalar

Table 2: Summary of Model Input and Outputs. Note the eclipse output rates are the rates which have been adjusted automatically to comply with wellbore hole pressure limits. They are therefore the average injection or withdrawal rate during the transition from time=t to time=t+1.

4.4.3 Data Normalisation

The spatial pressure values were scaled using min-max normalisation. The hydrogen saturation was not scaled due to values already being between zero and one. All input and output scalars were transformed to between zero and one using min-max normalisation, except for input rates as they were already in this range. For wellbore hole pressure, the minimum and maximum values were preset to 96.5 and 102, the wellbore hole pressure limits set in the eclipse simulation. Due to this, the idle wellbore hole pressure component is added as normalising the wellbore hole pressure with these values leads to errors. The idle wellbore hole component is one-hot encoded to indicate if it is an idle period explicitly.



Block	Input Shape	Output Shape	Description
1	[2, 12, 61, 61]	[64, 16, 64, 64]	Padding, 2 FiLM Conv
2	[64, 16, 64, 64]	[128, 8, 32, 32]	Max Pooling, 2 FiLM Conv
3	[128, 8, 32, 32]	[256, 4, 16, 16]	Max Pooling, 2 FiLM Conv
4	[256, 4, 16, 16]	[512, 2, 8, 8]	Max Pooling, 2 FiLM Conv
5	[512, 2, 8, 8]	[1024, 1, 4, 4]	FiLM Conv
6	[1024, 1, 4, 4]	[512, 2, 8, 8]	UpConv, 2 FiLM Conv
7	[512, 2, 8, 8]	[256, 4, 16, 16]	UpConv, 2 FiLM Conv
8	[256, 4, 16, 16]	[128, 8, 32, 32]	UpConv, 2 FiLM Conv
9	[128, 8, 32, 32]	[64, 16, 64, 64]	UpConv, 2 FiLM Conv, Conv3d
10	[64, 16, 64, 64]	[2, 12, 61, 61]	Crop to Spatial Output dim
11	[64, 16, 64, 64]	[64, 12, 7, 7]	Crop to block around well
12	[64, 12, 7, 7]	[16, 12, 7, 7]	2 Conv3D
13	[16, 12, 7, 7]	[16, 2, 2, 2]	Pooling
14	[16, 2, 2, 2]	[128, 1]	Flattening to 1D array
15	[128, 1]	[32, 1]	3 Linear layers
16	[32, 1]	[1]	1 Linear layer: Injection Rate
17	[32, 1]	[1]	1 Linear layer: Withdrawal Rate
18	[32, 1]	[1]	1 Linear layer: Hydrogen Stored

Figure 1: Model Architecture used for all predictions

4.5 Model Design and Evaluation Methods.

4.5.1 Model Hyperparameters and Architecture

The architecture of the model can be seen in Figure 1. The final model architecture presented was developed from a trial and error process. Feature-wise linear modulation and adaptive average pooling was added to the standard U-Net architecture. This was added and adapted from the research of Abdellatif et al. (2025) and Cui et al. (2025). Feature-wise linear modulation was added to every convolution block to give the scalar inputs a larger impact.

The U-Net architecture was used as it allows for large scale features to be retained through skip connections. Finer scalar features can be extracted through the gradual decrease in the dimensions in the encoder and decoder blocks. Other architectures were considered such as long short-term memory models which have been shown to be valuable for predicting withdrawal rate (Du et al., 2025). Long short-term memory models in this context allow information from previous months to be passed forward. However, this means the model user (i.e the hydrogen storage operator) must have information from previous cycles to make a prediction. By using the architecture in Figure 1 the user is only required to have the information for one point in time. Feature-wise linear modulation was also added due to the reduction in gas saturation error observed by Abdellatif et al. (2025).

The model was trained with a batch size of 32. Weight decay was applied to avoid overfitting and a cosine scheduler was used due to unstable training in earlier models. The loss function was a equally weighted combination due to the models 5 outputs. For the two spatial predictions relative loss function was used (Mao et al., 2025). The loss function used for the moles stored output was mean square error. For injection and withdrawal DenseLoss was used (Steininger et al., 2021). The DenseLoss was adapted to not account for values of zero. This was done as they are either idle phases or the alternate phase of injection or withdrawal.

4.5.2 Model Prediction and Evaluation

To evaluate the model performance, the R^2 and mean absolute percentage error were calculated for the scalar outputs. The mean absolute percentage error was calculated to allow for comparison between the scalars. The plume saturation error and pressure relative error were calculated for the hydrogen saturation and pressure spatial predictions (Wen et al., 2023). The mean error was also calculated for both spatial predictions at the final month of every site for each individual grid cell. These errors were all calculated using teacher-forced rollout; meaning the input to generate a prediction is from the ground-truth Eclipse simulation results.

Further evaluation was carried out by assessing the impact of predictions from autoregressive rollout. This method used some of the outputs of the previous timestep to generate the next prediction. The hydrogen stored, hydrogen saturation and pressure predictions from time=t were then used as input to predict at t=t+1. The other input variables such as injection and withdrawal rate and wellbore hole pressure were inputted using the ground truth variables. These are referred to as the conditioning variables. To evaluate these predictions absolute percentage error of all the model outputs was calculated.

The results for both methods were calculated exclusively on the unseen test dataset. The predictions are made by the model after being trained for 90 epochs where the validation loss plateaued. The cycle lengths and operational rates within the test set are summarised in Table 3.

This method would be highly valuable for an operator of a hydrogen storage site and user of the model. The autoregressive rollout would allow an operator to predict behaviour beyond a single timestep from one single sample of data.

	Cycle Length			**Rate**		
Phase	Maximum	Minimum	Median	Maximum	Minimum	Median
Injection	11	1	6	1.000	0.335	0.713
Withdrawal	10	1	5	0.840	0.007	0.372
Idle	6	1	2	X	X	X

Table 3: Summary statistics of cycle lengths and operational rates in the unseen test set

5 Results

5.1 Natural Gas

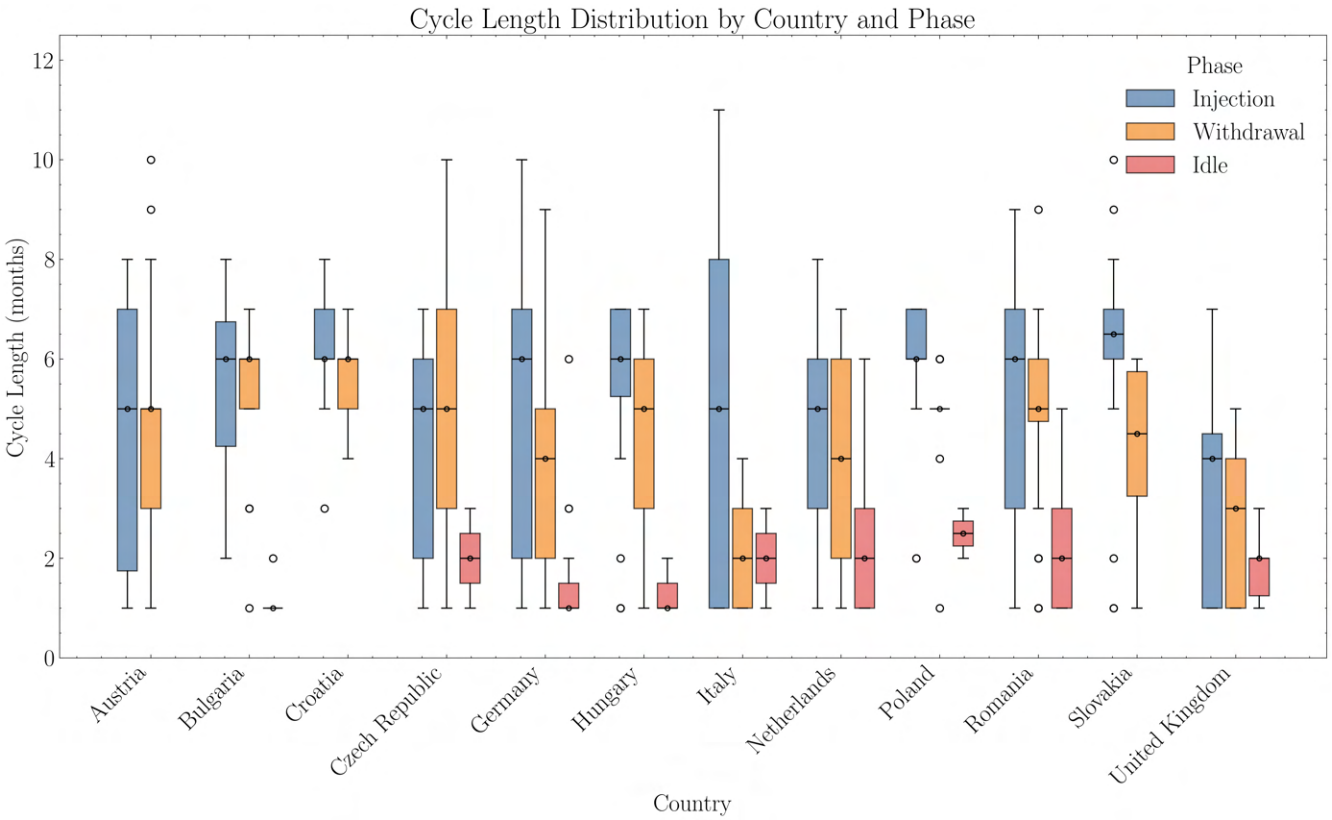
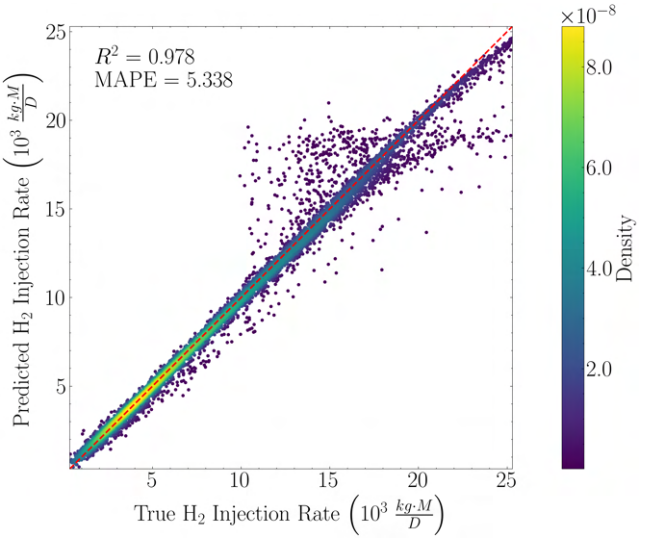


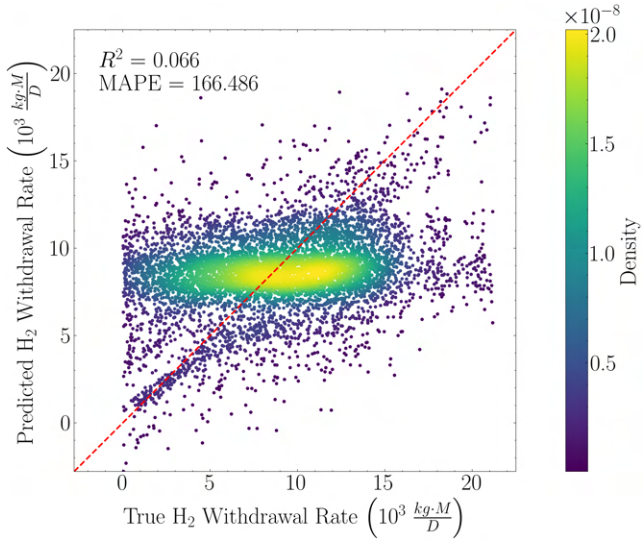
Figure 2: Distributions of injection, withdrawal and idle cycle lengths by Country

The lengths of the different cycle types vary significantly across the countries studied (fig 2). Injection cycles have the most variable length within a single country with the largest interquartile range, seen in Italy. Median injection cycle lengths have limited variability across all the countries. All the median cycle lengths are five or six months, with the exception of the United Kingdom. The median injection cycle length in the United Kingdom is the shortest across all countries. In Italy there is the largest disparity between the median injection and withdrawal cycle lengths, with the withdrawal median cycle length being the lowest. Withdrawal cycle lengths vary the most out of the cycle types across all the countries with the largest range of median cycle lengths. Withdrawal cycles are most likely to have anomalous cycle lengths indicated by the highest number of outliers. Idle phases are least variable with the largest interquartile range being 2 months, observed in the Netherlands and Romania. Idle cycles are not present in all countries.

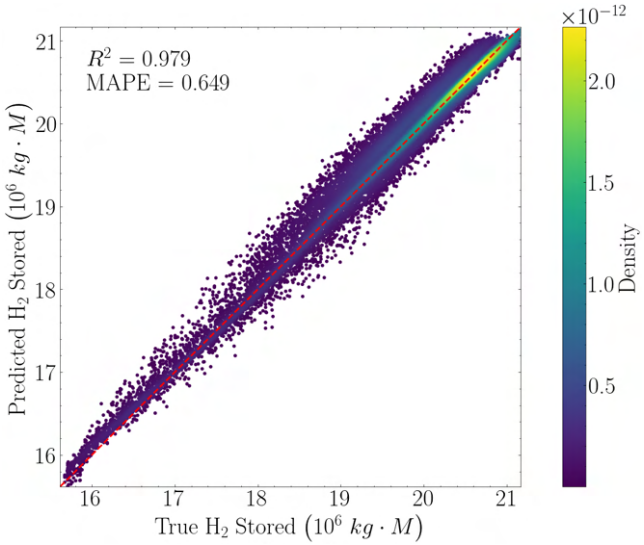
5.2 Prediction of Scalar Values



(a) True injection rate plotted against predicted rate.



(b) True withdrawal rate plotted against predicted rate.

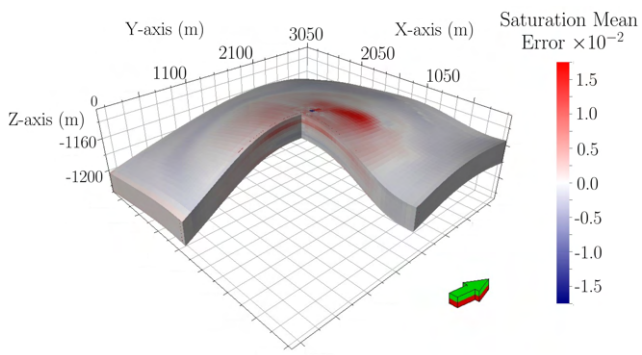


(c) True hydrogen stored plotted against predicted hydrogen stored.

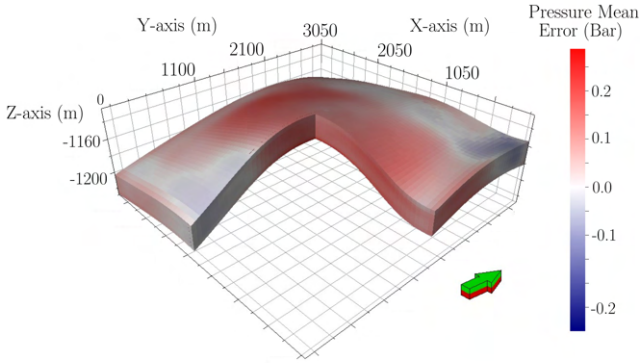
Figure 3: True vs predicted scatter plots for all scalar outputs. Note injection and withdrawal rates are displayed in kilogram-moles per day and hydrogen stored is kilogram-moles.

The model shows a varying ability to predict the three scalar variables (fig. 3). The true and predicted values for injection rate show strong alignment with an R^2 value of 0.978. For true injection rates greater than 10×10^3 $(kg \cdot M)/D$ the predictions become less accurate (fig. 3a). The model shows a poor ability to predict withdrawal rate with an R^2 value of 0.066. The mean absolute percentage reinforces the high inaccuracy of the model (fig. 3b). The model predicts the hydrogen stored with the highest accuracy. Despite a similar R^2 value to injection rate, the predictions do not significantly deviate from the perfect prediction line. The higher accuracy is highlighted most clearly by the mean absolute percentage error being 0.649 (fig. 3c).

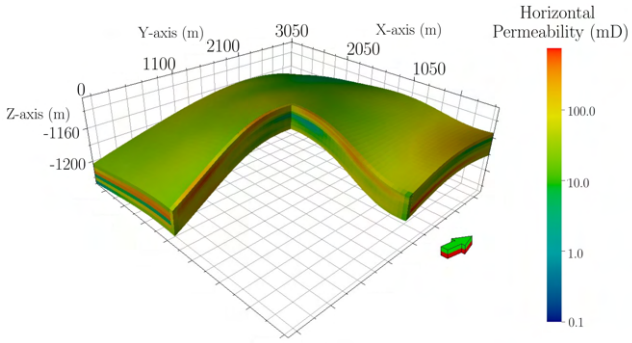
5.3 Spatial Prediction



(a) Saturation Prediction Errors



(b) Pressure Prediction Errors

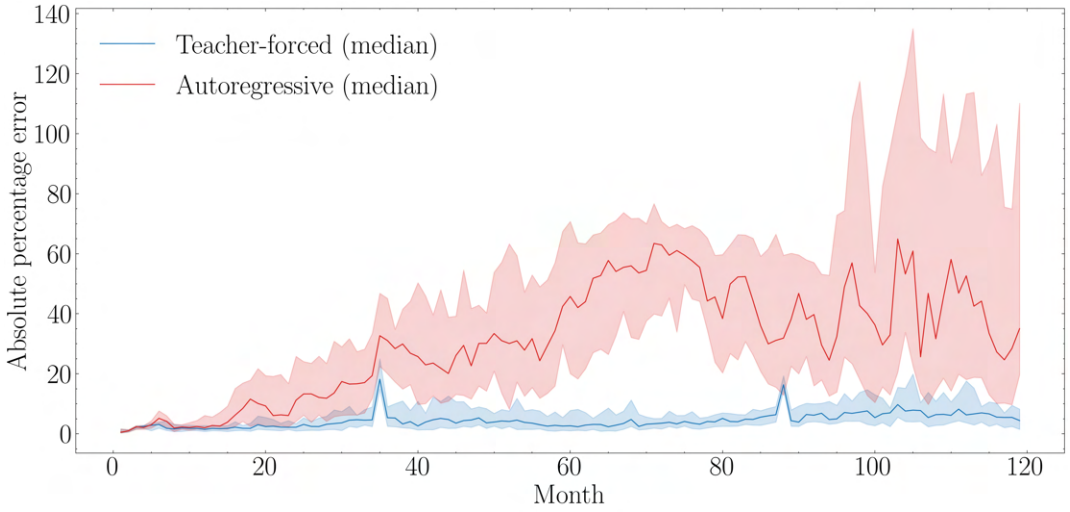


(c) Reservoir Horizontal Permeability

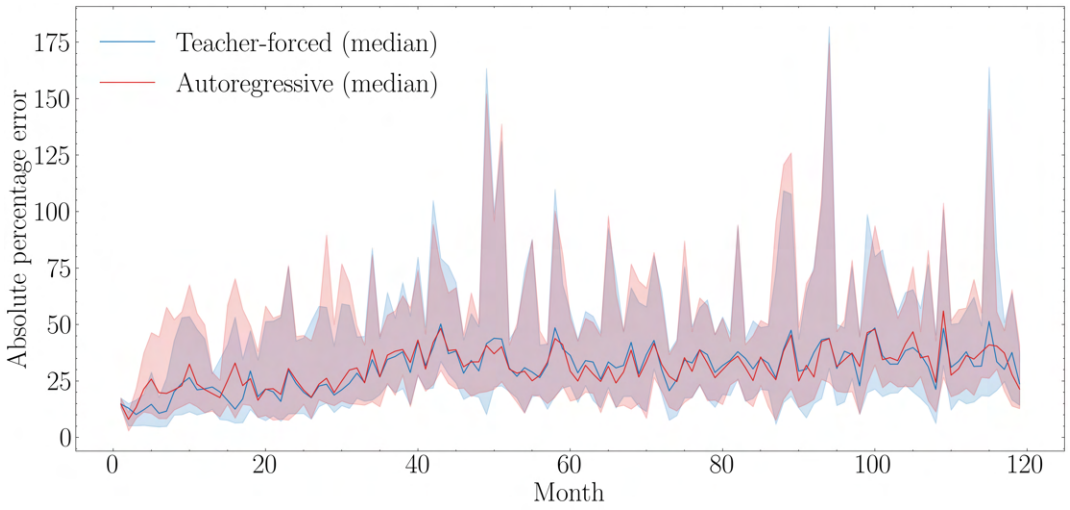
Figure 4: Diverging colourmap of spatial variables errors. Note the figures are done with a vertical exaggeration of 10.

The model's prediction of saturation is highly accurate with the maximum error being 1.6%. This is reinforced by the plume saturation error being 0.33%. The highest errors seen are closest to the well, overpredicting the saturation (fig. 4a). At the base of the anticline, saturation is underpredicted; however, the size of the underprediction is low. The model predicts the pressure with high accuracy. The errors are not as concentrated as the saturation errors. The maximum error of all grid cells in the test set is 0.26% and the pressure relative error is 0.14%. In the north-east corner of the reservoir, there is an area of underpredicted errors (Fig. 4b).

5.4 Autoregressive Rollout



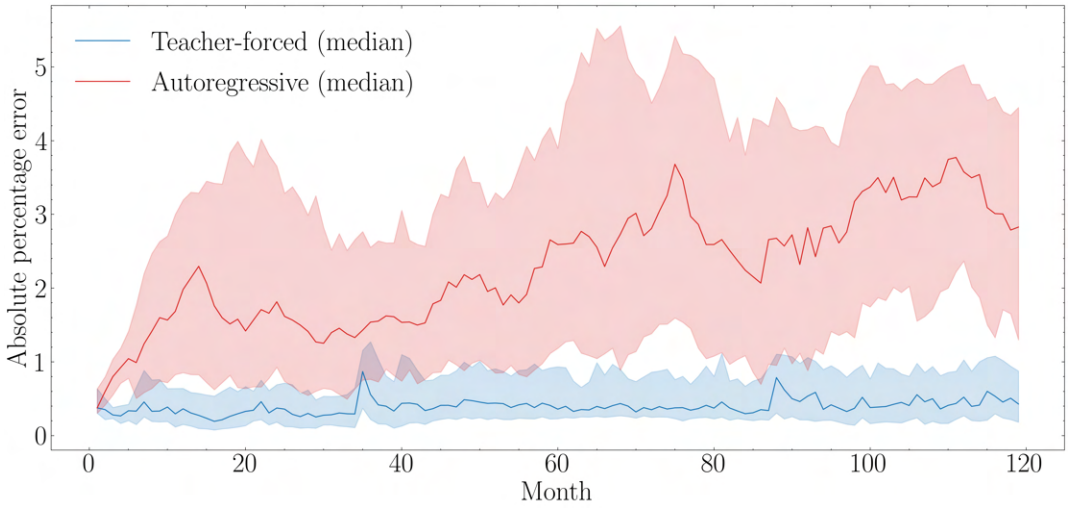
(a) Comparison of absolute percentage error between teacher-forced rollout (blue) and autoregressive rollout for injection rates (red).



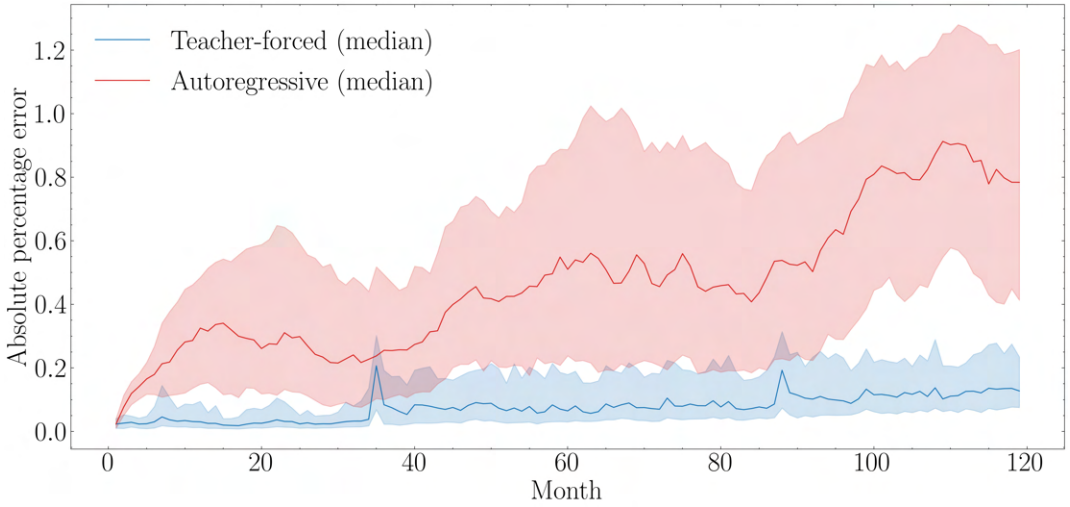
(b) Comparison of absolute percentage error between teacher-forced rollout (blue) and autoregressive rollout for withdrawal rates (red).

Figure 5: Comparison of errors between teacher-forced rollout and autoregressive rollout. Note the shaded areas in both images are the interquartile range.

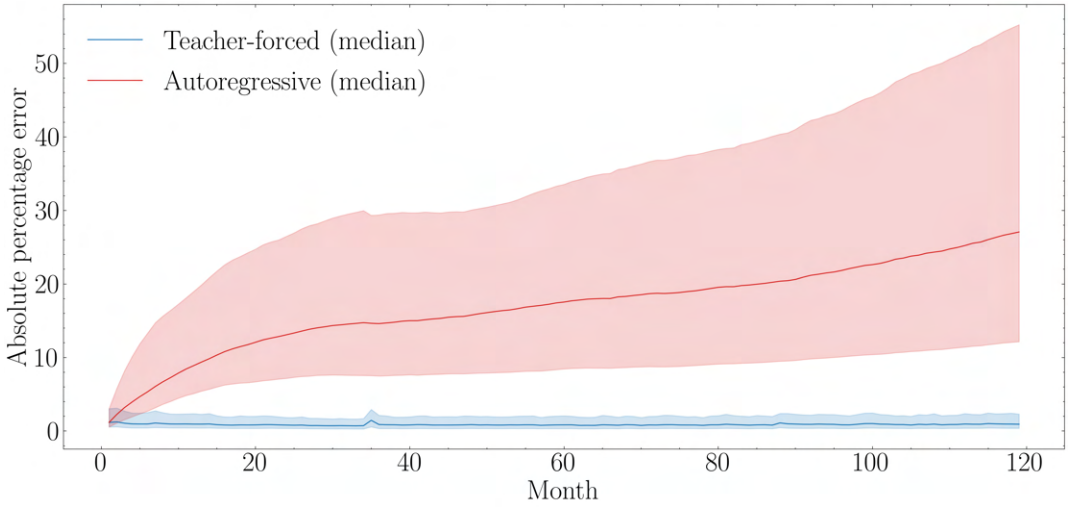
The results of the autoregressive rollout for injection and withdrawal rates show errors of the same order of magnitude as the teacher-forced rollout at a lower number of months (fig. 5). The autoregressive rollout injection predictions track the teacher-forced rollout with very small deviation up to month 10. The interquartile range of the autoregressive rollout predictions is a maximum absolute percentage error of 5%. After this period, the median increases to a peak before then decreasing and fluctuating with a constant amplitude. The interquartile range follows a similar trend before increasing to a maximum absolute percentage error of 100% when the median fluctuates (fig. 5a). The error of autoregressive rollout withdrawal rate predictions do not follow the same trend. The autoregressive rollout withdrawal predictions have approximately the same error as the teacher forced withdrawal rate predictions across the full study period (fig. 5b).



(a) Hydrogen stored prediction errors.



(b) Pressure prediction errors.



(c) Saturation prediction errors

Figure 6: Comparison of prediction errors from teacher forced (blue) and autoregressive (red) rollout over time. Note the error band in all images is the interquartile range.

The autoregressive rollout errors from hydrogen stored predictions show a non-linear trend which deviates significantly from the teacher-forced error predictions. The error increases initially before decreasing and then gradually rising. The error then decreases until month 90 before increasing again (fig 6a). The autoregressive rollout pressure prediction error follows a similar trend to the hydrogen stored error. However, at month 85 the trends diverge. The pressure error increases to its maximum which occurs at month 110 (fig. 6b). The interquartile range rapidly increases initially for both hydrogen stored and pressure variables; however, after this it remains constant. The saturation error does not follow this same trend. The saturation error median and interquartile range increase smoothly the entire time.(fig. 6c).

6 Discussion

6.1 Natural Gas Data

The results of the natural gas data analysis highlight the need for using varying cycle lengths in the development of surrogate models for underground hydrogen storage. The variations in cycle lengths across Europe are likely due to variations in the reservoir properties, market demands, operational models and varying seasons. The six month idle phases observed in the Netherlands are due to the site being at maximum capacity (GIE, 2026). The withdrawal cycle median being lowest in Italy is potentially due to having a shorter winter period. This is typically when gas is produced to meet increased market demand. Another explanation for the low median withdrawal cycle length is the reservoir properties of the Cornegliano gas storage site. This is the only site in Italy used in this study and it has the lowest total gas capacity of all sites (GIE, 2026). This may result in rapid depletion of the storage and shorter withdrawal cycle lengths. A final explanation is the site's "5+2" operational model during the withdrawal season in winter. This model means the site withdraws gas for five days during the week and injects gas for two days at the weekend (Lombardi et al., 2025). If injection outweighs the withdrawal amount, the withdrawal periods may appear much shorter. This operational model is designed to optimize storage operations. Seasonal variations across the different countries potentially explains the observation that the median injection cycle length is lowest in the United Kingdom. The summer months are generally when gas is injected. As a result, the United Kingdom having a shorter summer period would explain the shorter injection cycle lengths.

The results and analysis highlight the significant variations in cycle lengths and the multiple impacting variables. Due to this, training a deep learning surrogate would allow for rapid assessment of varying cycle lengths. The cyclic nature of underground hydrogen storage means deep learning models can be trained to assess these uncertainties once and then used multiple times. Although not clearly distinguishable in the results, this rapid assessment of uncertain cycle lengths would be invaluable for hydrogen storage operators when outlier events occur. Outlier events such as the invasion of Ukraine and Covid-19, can have a large impact on energy demand and assessing these impacts could be crucial.

6.2 Prediction of Scalar Values

The injection and withdrawal rate are important to predict. These are typically operator defined in natural gas storage and are a function of reservoir pressure and physical properties of the geological formation (Brkić et al., 2021). In underground gas storage minimum and maximum operating

pressures are defined. This is to ensure reservoir integrity and caprock tightness (Włodek et al., 2026).

In this study wellbore hole pressure limits were used to comply with this standard. As a result, the Eclipse input injection or withdrawal rate can differ compared to the Eclipse output rate. When the wellbore hole pressure in the eclipse simulation reaches the maximum or minimum pressures, the rates are automatically adjusted to ensure the limits are not breached. Therefore the model predicting the injection and withdrawal rates allows the adjusted rate to be predicted instantly.

In relation to use of the model by an operator of a hydrogen storage site, they can input an injection or withdrawal schedule and then receive the achievable injection or withdrawal rate. This allows hydrogen storage operators to forecast the amount of energy they can store and provide.

The prediction of injection rates is mostly very accurate. The accuracy drops above 10 $\times 10^3$ ($kg \cdot M/D$). This is likely due to their being less data points in this range, to limit this the DenseLoss was used (Steininger et al., 2021). The ability of the model to predict withdrawal rate is significantly lower than injection rate predictions. The R^2 value is only slightly better than predicting the mean. An explanation for this is that wellbore hole pressure changes much faster during withdrawal than injection which the model may struggle to learn (Appendix A1). To try and limit the impact of this, the three wellbore hole pressure components were added; however, this had a limited impact. The hydrogen stored component was added to the scalar inputs to try address this issue. The model predicts the amount of hydrogen stored with high accuracy. This is unexpected due to the models weak ability to predict withdrawal rate. The most plausible explanation is the difference in scales. The hydrogen stored prediction will not be affected significantly if the model learns to decrease during a withdrawal phase by the mean amount as it has much larger values. Despite the values being normalised, the difference in scales will have an impact. As the hydrogen stored is predicted independently, it will learn the withdrawal rate implicitly and the effects of it in relation to its own scale.

A hydrogen storage operator using the model can accurately predict the rate they will be able to store hydrogen and how much will be stored. This is crucial for forward planning. Accurately forecasting hydrogen deliverability will not be possible.

6.3 Spatial Prediction

The pressure predictions are highly accurate. Comparing the observation of underprediction in the north-east corner to the horizontal permeability, an explanation can be hypothesised. The horizontal permeability is higher than the surrounding area and this contrast may not be learnt by the model (fig. 4c). Overall, the pressure predictions, although errors appearing wide-spread, are highly accurate.

The saturation errors are highest closest to the well. This is likely due to it being where saturation will be most variable in injection and withdrawal phases. The observation of underprediction at the base of the anticline is potentially due to it being the mean plume front. Due to the hydrogen being driven upwards, the base of the anticline will have lower saturation. As the plume front will differ for all sites, the model does not learn to capture plume front physics entirely. There is also a layer of relatively high overpredictions by the well. When comparing to the horizontal permeability it can be seen that there is an area of lower permeability. The model therefore is unable to completely determine the difference in permeability and the effects of this.

The errors in both saturation prediction and pressure prediction could be reduced by inputting the reservoir properties into the model. However, this is unnecessary due to the high accuracy and the increased compute capacity it would require. The model can accurately predict the hydrogen plume shape and pressure within the reservoir. For hydrogen storage operators this is needed to avoid leakage of hydrogen and ensure reservoir integrity.

6.4 Autoregressive Rollout

The results of the injection rate autoregressive rollout predictions potentially give us an understanding of the key input variables to the model. The observation that the error increases initially loosely aligns with observations seen in the hydrogen stored and pressure autoregressive rollout results. This implies that these three variables receive and react to similar input signals resulting in errors propagating through these variables. The interquartile range significantly increasing after month 90 potentially indicates the results have gone significantly outside the model's training parameter space. The withdrawal rate autoregressive rollout prediction errors show no clear link to the autoregressive rollout inputs. This is not unexpected due to the results of the teacher-forced prediction being poor. This highlights that the model potentially does not receive enough information from the current inputs to make accurate predictions. Cumulative errors in the autoregressive rollout inputs do not propagate through to the withdrawal rate autoregressive rollout predictions.

The saturation autoregressive rollout prediction error does not follow a trend shown by other variables. This result implies it solely depends on the conditioning variables and saturation inputs. The saturation error increases smoothly which implies the prediction is not impacted by multiple variables. Another likely explanation is the size of the plume. In the first few months, the plume will be relatively small and throughout the time period it increases. Therefore, the error may grow due to the plume extent increasing and predictions becoming less accurate further from the well.

Overall, the model predictions from autoregressive rollout have a low error rate up to 15 months for injection rate, hydrogen stored and pressure. This means that operators can accurately predict and assess the impacts of injection cycles for the maximum cycle length observed in the natural gas data from one initial set of inputs. However, operators should be cautious in using the model to detect saturation leakage and forecasting deliverability.

6.5 Limitations and Directions for Future Work

The primary limitation of this study is the model used to generate the dataset. With no existing commercial operations of hydrogen storage in depleted gas fields, the accuracy of the model is not able to be verified. In addition to this, the resolution of the grid cells being 50m may not fully capture the dynamics of the hydrogen plume. Although there is no existing research on the resolution required for underground hydrogen storage, using carbon capture and storage as an analogue, resolutions of one to two metres are required (Wen et al., 2023).

Future work should primarily focus on improving the accuracy of predicting withdrawal rate, the key value for forecasting deliverability. One method to explore would be using a much finer temporal resolution. This would likely reduce the impact of large wellbore hole pressure changes, which is potentially causing the prediction to be inaccurate. Investigating the impacts of higher temporal resolution on autoregressive rollout predictions is also an interesting avenue to explore in future. A final, and most valuable, avenue for future research would be testing autoregressive

predictions on samples starting at different stages in the simulation and outside of the model's training time range. Achieving high accuracy on this is a very strong justification for developing deep learning based surrogate models. This would allow hydrogen storage operators from one single input to predict multiple future timesteps.

7 Conclusion

In this study we develop a surrogate model that uses varying cycle lengths and operational parameters. The surrogate model can accurately predict hydrogen saturation and pressure fields with maximum error rates of 1.6% and 0.26% respectively. Injection rate and hydrogen stored predictions are highly accurate with errors of 5.54% and 0.649%. Autoregressive rollout predictions are shown to be accurate for forecasting the injection rate, hydrogen stored and pressure for the maximum injection cycle length seen in the natural gas data. Compared to previous studies, this research assesses key operational parameters more thoroughly. The implications of this study are that: (1) underground hydrogen storage surrogate models need to use varying cycle lengths and operational rates if they are to be operated in a similar way to natural gas storage, (2) future operators have a framework to develop an accurate surrogate model which can forecast the impacts of different injection cycle lengths and rates and (3) future research has a proposed solution to improve prediction of hydrogen withdrawal rates and reduce compute required.

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A Results and Discussion

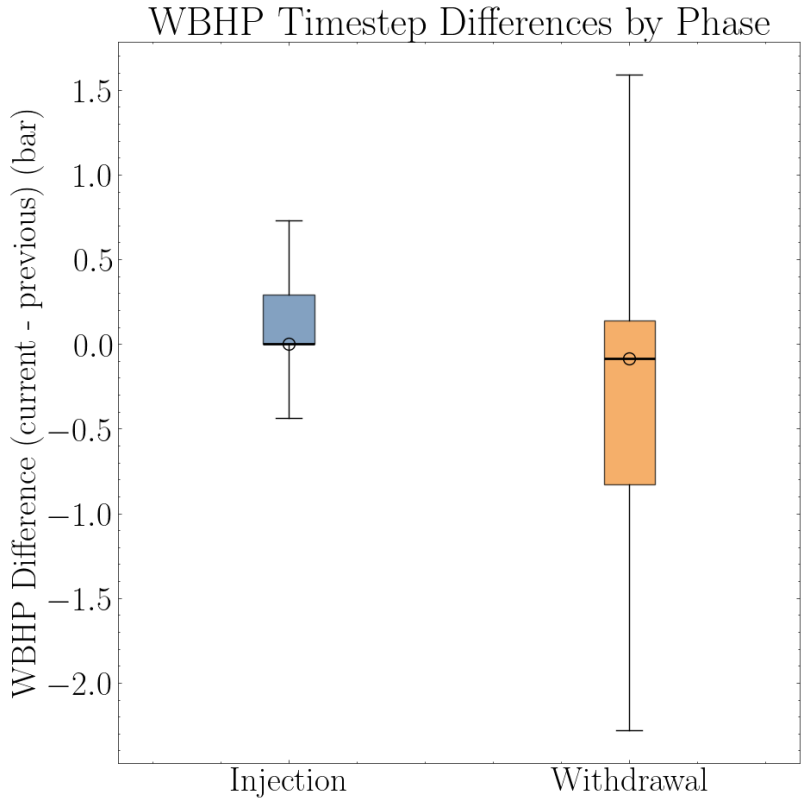


Figure A1: The distribution of wellbore hole pressure between current wellbore hole pressure and previous wellbore hole pressure, highlighting the difference during injection and withdrawal phases. Note the values only compare to values within the same cycle. This means the first month of each cycle is not included.